Parallel Computing Assignment

CS F422, Semester II 2024-25

Submitted by

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Problem A and B

Design aspects and
Optimizations:

1) Parallelly parsing JSON objects on GPU.
2) Privatization for adding ratings uses shared caches.
3) Efficient memory allocation for ASINs
4) Storing ratings as integers instead of strings or floats.

Problem C

86.21% reduction in execution time.

```
❯ time ./cuda_toprated
Processed 160052 products.
Timetaken for ratingsAdd: 1.982720
Timetaken for ratingsDivide: 0.010240
Index: 23, Asin: 1059998254, Value: 5.000000
Index: 52, Asin: 1465023879, Value: 5.000000
Index: 89, Asin: 6875467534, Value: 5.000000
Index: 117, Asin: 987604088X, Value: 5.000000
Index: 118, Asin: 9875918571, Value: 5.000000
Index: 122, Asin: 9875961817, Value: 5.000000
Index: 129, Asin: 9966635335, Value: 5.000000
Index: 134, Asin: 9966694242, Value: 5.000000
Index: 137, Asin: 9967221321, Value: 5.000000
Index: 148, Asin: 9966695516, Value: 5.000000
./cuda_toprated  281.47s user 1.38s system 99% cpu 4:43.84 total
```

Unoptimized

```
❯ time ./cuda_toprated_optimized
Allocating 4186359340 bytes for string data
Timetaken for parsing JSON in GPU: 35.279873
Processed 160052 products.
Timetaken for ratingsAdd: 1.741632
Timetaken for ratingsDivide: 0.028672
Index: 23, Asin: 1059998254, Value: 5.000000
Index: 52, Asin: 1465023879, Value: 5.000000
Index: 89, Asin: 6875467534, Value: 5.000000
Index: 117, Asin: 987604088X, Value: 5.000000
Index: 118, Asin: 9875918571, Value: 5.000000
Index: 122, Asin: 9875961817, Value: 5.000000
Index: 129, Asin: 9966635335, Value: 5.000000
Index: 134, Asin: 9966694242, Value: 5.000000
Index: 137, Asin: 9967221321, Value: 5.000000
Index: 148, Asin: 9966695516, Value: 5.000000
./cuda_toprated_optimized  36.45s user 2.68s system 99% cpu 39.274 total
```

Optimized

Problem D and E

Design aspects and optimizations-

1) Sentiment scores are aggregated on the GPU.
2) Data preprocessing takes place on the CPU, to minimize data transfer.
3) Words are stored using an unordered_map on the CPU.
4) Reviews with no matching words in the lexicon are skipped.

Problem F

```
❯ time ./cuda_reviewanalysis
Processed 7506 lexicons
Timetaken for scoresAccumulate: 8.429568
Total no. of reviews: 5792570
Positive: 4998501 Negative: 780644 Neutral: 13425
./cuda_reviewanalysis  388.98s user 1.90s system 99% cpu 6:33.24 total
```

Unoptimized

```
❯ time ./cuda_reviewanalysis_optimized
Processed 7506 lexicons
Timetaken for scoresAccumulate: 11.804672
Timetaken for scoresCount: 1.425408
Total no. of reviews: 5792570
Positive: 4998501 Negative: 780644 Neutral: 13425
./cuda_reviewanalysis_optimized  389.84s user 1.22s system 99% cpu 6:32.60 total
```

Optimized

Problem G and H

Design aspects and optimizations-

1) Review Counts are stored using an unordered map.
2) Race conditions are avoided by using thread-local storage.
3) The JSON data is parsed in parallel using OpenMP.
4) Data is loaded into memory before processing to reduce I/O overhead.

Problem I

83.95% reduction in execution time.

```
❯ time ./c_elaborate
514782
./c_elaborate  288.71s user 1.63s system 99% cpu 4:51.99 total
```

Unoptimized

```
❯ time ./ompgpu
514782
./ompgpu  604.89s user 2.97s system 1301% cpu 46.706 total
```

Optimized